

1. Chemical identity

<i>Chemical name:</i> HP-BIOACTIVA		<i>Chemical classification:</i> Biological powder
<i>Synonyms:</i> Bioadditive, Bioinoculant		<i>Trade name:</i> HP-ctiva Bioa
<i>Formula:</i> Microbial-nutrient Blend		
<i>Regulated identification</i>	<i>Shipping name:</i> HP-Bioactiva	
<i>Codes/Label:</i>	<i>Hazchem code:</i> class 4	
	<i>Hazardous waste ID No:</i> N.A.	
<i>Hazardous ingredients</i> : Microorganisms		
HP-Bioactiva is a complex blend of microorganisms and nutrients. The exact composition of the product may depend upon the microbe: nutrient (1:1) blend.		

2. Physical and chemical data

<i>Boiling point/Range (deg.C) : NA</i>	<i>Physical state: Powder</i>	<i>Appearance: Yellowish</i>
<i>Melting/freezing point (deg.C) : NA.</i>		
<i>Vapor pressure: NA</i>		
<i>Odor: Protein like odor</i>		
<i>solubility in water : @ 30 deg.C: 100 %</i>		
<i>Specific Gravity : NA.</i>	<i>ph :5.5-7</i>	
<i>Others: NA</i>		

3. Fire and explosion Hazard data

<i>Flammability:</i> Non Flammable.		
<i>Flash point(deg C) :</i> NA		
<i>TDG Flammability:</i> NA		
<i>Ignition Temp (deg C) :</i> NA		
<i>Explosion sensitivity to impact:</i> not sensitive to <i>Explosion sensitivity to static electricity:</i> not sen		
<i>Hazardous Combustion Products:</i> CO ₂		
<i>Hazardous Polymerization:</i> N.A. Mechanical Impact sitive to static electricity		
<i>Combustible powder:</i> NA	<i>Explosive material:</i> No	<i>Corrosive material:</i> No

4. Reactivity data

<i>Chemical stability:</i> Stable under cool conditions (18-22°C) for 6 months and away from sunlight and moisture.
<i>Hazardous reaction products:</i> on fire it will decompose and generate CO ₂

5. Health Hazard data

Routes of entry: Inhalation, Skin absorption, ingestion *Effects*

of Exposure/ symptoms:

Inhalation: Inhalation may cause sneezing, headache, nausea, vomiting, allergic reaction

Skin Contact: Skin-dryness, cracking, irritation eyes watering.

Emergency treatment:

In case of contact with Skin, flush with excess fresh water, arts with clean water and seek medical assistance.

In case of excessive inhalation, wash nasal p for microbial infection.

Permissible Exposure limit: *L.D50 (Oral-Rat Odor threshold: N.A.)* : NA. *L.C 50:* (rat for 4hrs): NA

6. Preventive measures

Personal Protective equipment: Gloves, eye protection preferred.

Handling and storage precautions: Store under cool conditions (AC) and away from direct exposure to sunlight and moisture

7. Emergency and first aid measures

Fire:

Fire extinguishing media: Water may be used to cool fire-exposed containers.

Exposure: *First aid measures:*

Skin contact ; In case of contact with Skin flush with fresh water

Inhalation: In case of excessive inhalation, flush parts with water and move the victim to fresh air. Seek medical assistance in case of allergic reactions.

Ingestion: Rinse mouth with clean water. Seek medical assistance.

Antidotes/Dosages: N.A.

Spills:

Steps to be taken: Clean with cloth, wipe the area with common floor disinfectant.

Waste Disposal method: N.A.

8. Additional Information/Reference

9. Manufacture/Suppliers Data

Manufacture (Name of Firm) : Hindustan Petroleum Corporation Limited.

Supplier/dealer Data.

Name: HPCL, Bangalore

Mailing address:

Contact Persons: Dr. Harshad Velankar

10. DISCLAIMER

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